Representations of SL_n & $\mathfrak{sl}_n$

1) Harish-Chandra (HC) isomorphism for the center of $\mathcal{U}(\mathfrak{g})$.

1.0) Intro.

$\mathbb{F}$: alg. closed char 0 field, $\mathfrak{g} = \mathfrak{sl}_n(\mathbb{F})$, $\mathcal{Z} := Z(\mathcal{U}(\mathfrak{g}))$.

Goal: Describe the algebra $\mathcal{Z}$ and understand its action on $\Delta(\lambda)$, and its unique irred. quotient $L(\lambda)$ ($\lambda \in \mathfrak{h}^*$). Apply this description to prove that every finite dimensional $\mathfrak{g}$-representation is completely reducible.

1.1) Homomorphism $\mathcal{Z} \longrightarrow \mathcal{U}(\mathfrak{h})$.

To describe $\mathcal{Z}$ we construct an algebra homomorphism $\mathcal{Z} \to \mathcal{U}(\mathfrak{h})$. Later we'll see it's injective and describe the image, hence describing $\mathcal{Z}$. This homomorphism will also be used to describe how $\mathcal{Z}$ acts on $\Delta(\lambda)$.

Recall: for $\beta = \varepsilon_i - \varepsilon_j$ $(i<j)$ we write $f_\beta := E_{ji}$, $e_\beta = E_{ij}$. For $i = 1, \dots, n-1$, $h_i := E_{ii} - E_{i+1,i+1}$; $N = \frac{n(n-1)}{2}$, $\beta_1, \dots, \beta_N$ - all positive roots.

PBW Thm: $\mathcal{U}\mathfrak{g}$ has basis $f^{\vec{k}} h^{\vec{\ell}} e^{\vec{m}}$ $(\vec{k}, \vec{m} \in \mathbb{Z}^N_{\geq 0}, \vec{\ell} \in \mathbb{Z}^{n-1}_{\geq 0})$ (1)

$\mathfrak{g}$, hence $\mathfrak{h}$, acts on $\mathcal{U}(\mathfrak{g})$ by ad: $\operatorname{ad}(x)a := [x,a]$ $(x \in \mathfrak{h}, a \in \mathcal{U}\mathfrak{g})$.

Exercise: (1) is a weight vector of weight $\sum_{j=1}^{N} (m_j - k_j)\beta_j$ (hint: $\forall x \in \mathfrak{h}$, $a, b \in \mathcal{U}(\mathfrak{g})$, have $[x, ab] = [x,a]b + a[x,b]$).

Now we define the map $\pi : U(\mathfrak{g}) \to U(\mathfrak{h})$ sending $a \in U(\mathfrak{g})$ to the sum of all monomials in the expansion of a in (1) that only have h_i's.

Example: for $C = \frac{1}{2}h^2 + h + 2fe \in Z \subset U(\mathfrak{sl}_2) \Rightarrow \pi(C) = \frac{1}{2}h^2 + h$.

Note that all monomials in the expansion of $a - \pi(a)$ must have $k_j > 0$, $m_{j'} > 0$ for some j, j'. If a has weight 0 ($\Leftrightarrow [\mathfrak{h}, a] = 0 \Leftarrow a \in Z$), then $\forall$ monomial in a must have weight 0. So, $\pi(a) \in U(\mathfrak{h})$ satisfies

$$a = \pi(a) + \sum_{j=1}^{N} ? e_{\beta_j}. \tag{2}$$

For $z \in Z$, set $HC_z := \pi(z)$. Note that $\mathfrak{h}$ is an abelian Lie algebra $\Rightarrow U(\mathfrak{h}) = S(\mathfrak{h}) = \mathbb{F}[\mathfrak{h}^*]$. So we view HC_z as a polynomial on $\mathfrak{h}^*$.

Proposition: 1) $\forall\ z \in Z, \lambda \in \mathfrak{h}^*$, z acts on $\Delta(\lambda)$ & $L(\lambda)$ by $HC_z(\lambda)$.

2) $z \mapsto HC_z$ is an algebra homomorphism $Z \to U(\mathfrak{h})$.

Proof: 1) Since $e_\beta v_\lambda = 0$ $\forall$ positive root β, (2) $\Rightarrow z v_\lambda = HC_z(\lambda) v_\lambda$; $\forall v \in \Delta(\lambda)$ $\exists\, a \in U(\mathfrak{g}) \mid v = a v_\lambda$. Have $zv = z a v_\lambda = a z v_\lambda = HC_z(\lambda) a v_\lambda = HC_z(\lambda) v$.

The claim for $L(\lambda)$ follows b/c $\Delta(\lambda) \twoheadrightarrow L(\lambda)$.

2) $z \mapsto HC_z$ is $\mathbb{F}$-linear by construction. By 1), $HC_{z_1 z_2}(\lambda) = HC_{z_1}(\lambda) HC_{z_2}(\lambda)$ $\forall\ \lambda \in \mathfrak{h}^*$, $z_1, z_2 \in Z$. Indeed both sides are the scalars by which $z_1 z_2$ acts on $\Delta(\lambda)$ computed in two different ways.

So $HC_{z_1 z_2} = HC_{z_1} HC_{z_2}$. $\square$

1.2) Harish-Chandra isomorphism.

For $i=1,\dots,n-1$, define $s_i\cdot\lambda=\lambda-(\langle\lambda,h_i\rangle+1)\alpha_i$ so that $s_i\cdot$ is an affine map $\mathfrak{h}^*\to\mathfrak{h}^*$.

Proposition: $\forall z\in Z, \lambda\in\mathfrak{h}^*$ have $HC_z(\lambda)=HC_z(s_i\cdot\lambda)$.

Proof: Case 1: $\langle\lambda,h_i\rangle\in\mathbb{Z}_{\geqslant 0}$. By Sec 1.1 of Lec 15, $\exists$ nonzero $\mathcal{U}(\mathfrak{g})$-linear homomorphism $\Delta(s_i\cdot\lambda)\to\Delta(\lambda)\Rightarrow$ scalars of actions of $z\in\mathcal{U}(\mathfrak{g})$ on $\Delta(s_i\cdot\lambda),\Delta(\lambda)$ coincide. By Prop 1, $HC_z(\lambda)=HC_z(s_i\cdot\lambda)$.

Case 2: general. The locus $\{\lambda\in\mathfrak{h}^*\mid\langle\lambda,h_i\rangle\in\mathbb{Z}_{\geqslant 0}\}$ is a countable union of hyperplanes: $\{\lambda\in\mathfrak{h}^*\mid\langle\lambda,h_i\rangle=m\}$ for $m\in\mathbb{Z}_{\geqslant 0}$. Any polynomial vanishing on such locus is identically 0. Apply this to the polynomial $\lambda\mapsto HC_z(\lambda)-HC_z(s_i\cdot\lambda)$ & finish the proof. $\square$

Example: For $\mathfrak{sl}_2$: $\mathfrak{h}\simeq\mathbb{C}$ w. $h\leftrightarrow 1\rightsquigarrow\mathfrak{h}^*\simeq\mathbb{C}$ w. $\alpha=2,\ \rho=1,\ s\cdot\lambda=-\lambda-2$. Since $HC_C=\frac{1}{2}h^2+h$, we get $HC_C(\lambda)=\frac{1}{2}\lambda^2+\lambda=HC_C(-\lambda-2)$.

In fact, $\lambda\mapsto s_i\cdot\lambda$ extends to an action of the Weyl group $W(=S_n)$ on $\mathfrak{h}^*$. Set $\rho=\frac{1}{2}\sum_{i<j}(\varepsilon_i-\varepsilon_j)=\sum_{i=1}^{n}\left(\frac{n+1}{2}-i\right)\varepsilon_i\in\mathfrak{h}^*$ so that $\langle\rho,h_i\rangle=1$ $\Rightarrow s_i\rho=\rho-\alpha_i$. Then $s_i(\lambda+\rho)-\rho=\lambda+\rho-\langle\lambda+\rho,h_i\rangle\alpha_i-\rho=\lambda-(\langle\lambda,h_i\rangle+1)\alpha_i=s_i\cdot\lambda$. Here $\alpha_i=\varepsilon_i-\varepsilon_{i+1}$ is a simple root.

Definition: The shifted action of W on $\mathfrak{h}^*$ is given by $w\cdot\lambda:=w(\lambda+\rho)-\rho$.

Consider the subalgebra $\mathbb{F}[\mathfrak{h}^*]^{(W,\cdot)}=\{f\in\mathbb{F}[\mathfrak{h}^*]\mid f(w\cdot\lambda)=f(\lambda),\ \forall\lambda\in\mathfrak{h}^*, w\in W\}$ of invariant polynomials. Since the elements s_i generate W, Proposition above implies $HC_z \in \mathbb{F}[\mathfrak{h}^*]^{(W,\cdot)}$ $\forall z \in Z$. The following will be proved next time.

Thm (Harish-Chandra) $z \mapsto HC_z : Z \xrightarrow{\sim} \mathbb{F}[\mathfrak{h}^*]^{(W,\cdot)}$.

Corollary: For $\lambda, \mu \in \mathfrak{h}^*$ TFAE

(1) $\lambda \in W \cdot \mu$.

(2) $HC_z(\lambda) = HC_z(\mu)$, $\forall z \in Z$.

Proof: (1) $\Rightarrow$ (2) is a direct consequence of the theorem. (2) $\Rightarrow$ (1) becomes: if $f(\lambda) = f(\mu)$ $\forall f \in \mathbb{F}[\mathfrak{h}^*]^{(W,\cdot)}$, then $\lambda \in W \cdot \mu$. This is exercise (hint: find a polynomial f that is 1 on $W \cdot \lambda$, 0 on $W \cdot \mu$ and average w.r.t. W-action: $f \mapsto \frac{1}{|W|} \sum_{w \in W} f(w \cdot ?)$).

1.4) Algebra $\mathbb{F}[\mathfrak{h}^*]^{(W,\cdot)}$

Consider the affine isomorphism $\tau : \mathfrak{h}^* \xrightarrow{\sim} \mathfrak{h}^*$, $\lambda \mapsto \lambda + \rho$ so that $\tau(w \cdot \lambda) = w\,\tau(\lambda)$. So τ gives rise to an isomorphism $\tau : \mathbb{F}[\mathfrak{h}^*]^{(W,\cdot)} \xrightarrow{\sim} \mathbb{F}[\mathfrak{h}^*]^W$. Let's describe the target. Embed $\mathfrak{h}^* \hookrightarrow \mathbb{F}^n$ via $\varepsilon_i \mapsto e_i - \frac{1}{n}(e_1 + \ldots + e_n)$ w. image $= \{(x_1, \ldots x_n) \mid x_1 + \ldots + x_n = 0\}$. Let $\sigma_i \in \mathbb{F}[\mathbb{F}^n]$ be the ith elementary symmetric polynomial: $\sigma_i(x_1, \ldots x_n) = \sum_{j_1 < \ldots < j_i} x_{j_1} \ldots x_{j_i}$.

Lemma: $\mathbb{F}[\mathfrak{h}^*]^W$ is the algebra of polynomials in $\sigma_2, \ldots, \sigma_n$.

Proof: exercise (hint: show $\mathbb{F}[\mathfrak{h}^*]^W = \mathbb{F}[\mathbb{F}^n]^W / (\sigma_1)$ & use the fundamental thm about symmetric polynomials to describe $\mathbb{F}[\mathbb{F}^n]^W$. $\square$

Exercise: $Z \subset \mathcal{U}(\mathfrak{sl}_2)$ is generated by C.

1.4) Application: complete reducibility.

Thm: Every finite dimensional representation of $\mathfrak{g}$ is completely reducible.

Lem: Let $\lambda \neq \mu \in \Lambda_+$. Then $HC_z(\lambda) \neq HC_z(\mu)$ for some $z \in Z$.

Proof: The entries of $\lambda+\rho, \mu+\rho$ are still decreasing so $\lambda \in W\cdot\mu$ ($\Leftrightarrow$ $\lambda+\rho \in W(\mu+\rho)$ for the usual action $\Leftrightarrow$ $\lambda+\rho$ is obtained from $\mu+\rho$ by permutation) implies $\lambda=\mu$. So, thx to Corollary in Sec 1.2, $\exists\, z \in Z$ acting on $L(\lambda), L(\mu)$ by different scalars. $\square$

Sketch of proof of Thm: Let $\chi: Z \to \mathbb{F}$ be a homomorphism. E.g. for $\lambda \in \mathfrak{h}^*$ we can consider $\chi_\lambda: z \mapsto HC_z(\lambda)$. For a $\mathfrak{g}$-rep. V & $\chi: Z \to \mathbb{F}$ set $V^\chi = \{v \in V \mid \forall\, z \in Z\ \exists\, n_z \in \mathbb{Z}_{>0} \text{ s.t. } (z-\chi(z))^{n_z} v = 0\}$ - the intersection of generalized e-spaces. Note that if $z_2 \ldots z_n \in Z$ correspond to $\sigma_2 \ldots \sigma_n \in \mathbb{F}[\mathfrak{h}^*]^W$, then V^χ is the intersection of the generalized e-spaces $V_{\chi(z_i)}(z_i)$, $i = 2, \ldots n$. Note that since Z is the center, V^χ is a $\mathfrak{g}$-subrep $\forall\, \chi$.

Exercise: If V is finite dimensional, then

1) $V = \bigoplus_{\chi: Z \to \mathbb{F}} V^\chi$

2) $V^\chi = 0$ unless $\chi = \chi_\lambda$ for some $\lambda \in \Lambda_+$.

So it's enough to show that V is completely reducible if $V = V^{\chi_\lambda}$.

First observe that if $v \in V_\mu$ satisfies $e_\beta v = 0$ $\forall$ positive roots $\beta \Rightarrow \mu = \lambda$. if $v \in V_\mu$, then $zv = HC_z(\mu)v$ $\forall z \in Z$. This implies $HC_z(\mu) = HC_z(\lambda)$ $\forall z \in Z$ and, by Lemma, $\lambda = \mu$.

Now let $v \in V_\lambda \rightsquigarrow \varphi: \Delta(\lambda) \to V$ w. $v_\lambda \mapsto v$. Then $V' := \Delta(\lambda)/\ker\varphi$ is finite dimensional. We claim $V' \simeq L(\lambda)$. Indeed, since every highest weight of $\ker[V' \twoheadrightarrow L(\lambda)] \subset V$ has to be λ by the previous paragraph. On the other hand, all weights of $\ker[\Delta(\lambda) \twoheadrightarrow L(\lambda)]$ & hence of $\ker[V' \twoheadrightarrow L(\lambda)]$ are $< \lambda$. Combining these two observations show $V' \simeq L(\lambda)$.

Now the proof repeats the case of $\mathfrak{sl}_2$ (Sec 2 of Lec 6) & is left as an exercise. $\square$

Corollary: Every nonzero finite dimensional quotient of a Verma module is irreducible. In particular, $\tilde{L}(\lambda) \xrightarrow{\sim} L(\lambda)$ ($\tilde{L}(\lambda)$ was defined in Lec 15).

Proof: exercise.

Rem: We don't need the full power of HC isomorphism to prove the complete reducibility - there are more elementary proofs. We will use the theorem when we compute the character of $L(\lambda)$, $\lambda \in \Lambda_+$.

2) Proof, started.

2.1) $\mathcal{Z}$ vs $\mathcal{U}(\mathfrak{g})^G$.

To establish the HC isomorphism, we'll need an alternative description of $\mathcal{Z}$. Let G be a connected algebraic group w. Lie algebra $\mathfrak{g}$. Recall, Sec 1.2 of Lec 10, that G acts on $\mathcal{U}(\mathfrak{g})$ by algebra automorphisms $\rightsquigarrow$ the subalgebra $\mathcal{U}(\mathfrak{g})^G \subset \mathcal{U}(\mathfrak{g})$ of invariants.

Lemma: $\mathcal{Z} = \mathcal{U}(\mathfrak{g})^G$.

Proof: $\mathcal{Z} = \{a \in \mathcal{U}(\mathfrak{g}) \mid ad(x)a = 0\ \forall x \in \mathfrak{g}\}$. We write $\mathbb{F}$ for the trivial representation of $\mathfrak{g}$ or of G. Then

$$\begin{array}{ccl}
\mathcal{Z} & \ni & \varphi(1) \\
\sim\downarrow & & \updownarrow \\
\mathrm{Hom}_{\mathfrak{g}}(\mathbb{F}, \mathcal{U}(\mathfrak{g})) & \ni & \varphi \\
\| & \longleftarrow & \text{By Thm 2 in Sec 1.3 of Lec 7.} \\
\mathrm{Hom}_G(\mathbb{F}, \mathcal{U}(\mathfrak{g})) & \ni & \varphi \\
\sim\downarrow & & \updownarrow \\
\mathcal{U}(\mathfrak{g})^G & \ni & \varphi(1)
\end{array}$$

$\square$

3) Complements.

Here are some details for proving Theorem in Sec 1.4.

• Decomposition into "infinitesimal blocks": Let V be a $\mathfrak{g}$-representation (not necessarily finite dimensional). Let $\chi: \mathcal{Z} \to \mathbb{F}$ be an algebra homomorphism. Set

$$V^\chi = \{v \in V \mid \forall z \in Z\ \exists m > 0 \text{ s.t. } (z - \chi(z))^m v = 0\}$$

This is a $U(\mathfrak{g})$-submodule in V. If V is finite dimensional, then $V = \bigoplus_\chi V^\chi$. Moreover, z acts by $\chi(z)$ on every irreducible constituent of V^χ. It follows that $\chi(z) = HC_z(\lambda)$ for some $\lambda \in \Lambda^+$ whenever $V^\chi \neq \{0\}$. Moreover, by the observation in the proof of the theorem in Sec 1.4, in this case $L(\lambda)$ is the unique irreducible constituent of V^χ.

So assume $V = V^\chi \Leftrightarrow V$ is filtered by $L(\lambda)$ w. $\chi(z) = HC_z(\lambda)$. Then $L(\lambda) \otimes V_\lambda \xrightarrow{\sim} V$, the proof repeats that in Sec 1.3 of Lec 9. Details are left as an exercise.